What is this book?

This book, *From Lab to Laptop: Case Studies in Teaching Practical Courses Online*, provides a series of case studies which explore how higher education students and staff in a variety of international contexts navigated the transition of bringing practical, hands-on learning into an online environment. Although many publications exist on the topic of online learning, there are very few that specifically focus on how to teach practical elements of courses online. By sharing the experiences and reflections of this varied group of authors, the hope is to provide much-needed insights on this under-explored area of higher education pedagogy.

Why is the teaching of practical courses online an important topic?

When it comes to adapting learning for online environments, some courses are more challenging to adapt than others. Disciplines with substantial practical elements often require more consideration and creativity to be successfully redesigned for an online setting. This book includes case studies from a range of disciplines such as engineering, physiotherapy, physiology, chemistry labs, performing arts, business negotiation skills, media production, nursing, dance, healthcare, open education, educational development, agronomy and nutrition. These courses all include practical, hands-on learning that traditionally requires physical presence and is not easy to replicate online. For example, when teaching dance performance online, how should we provide learners with the opportunity to practise their performance skills and get feedback from their instructor? See Chapter 9 for a discussion of this. Or when teaching nutrition online, how do we provide learners with practical experience via work placements? Chapter 14 provides some suggestions. Precisely because these teaching scenarios present unique challenges, it is important that we share solutions and innovative practice. This book aims to address this need by providing a range of case studies from students and staff in different higher education contexts on how practical courses can be translated into online learning environments.

Online learning and blended learning are of increasing importance in the world of higher education. The onset of the Covid-19 pandemic in 2019-2020 expedited the shift towards these modalities, as it forced the global community to rapidly switch from in-person education to emergency online distance learning (Hodges et al., 2020). For example, the Higher Education Policy Institute in the UK reported that the number of students participating in online learning rose significantly during the pandemic (Neves & Hewitt, 2020) and in the USA 75% of university students transitioned to taking university courses online during that time (Hamilton & Swanston, 2024). In most cases, this resulted in teaching sessions being delivered using online meeting software, such as Zoom or Microsoft Teams, with an increased reliance on asynchronous tasks and digital resources.

This transition was not an easy one. As you will see from the chapters in this publication, there were a wide range of responses from both staff and students. Many struggled with adapting and found the process of online learning very isolating. Students, in particular, struggled with the mental health aspects of this emergency transition to online teaching. The potential repercussions on mental health are reflected by broader evidence across the sector. According to the Office for National Statistics in the UK, in the 2020-21 academic year 63% of students indicated that their well-being and mental health had worsened (Office for National Statistics, 2021).

On the other hand, since the pandemic, levels of satisfaction with online learning have significantly increased. A survey by Jisc indicates that 80% of students provided positive ratings on the quality of online learning on their course (Jisc, 2023). Although many programmes reverted to in-person instruction after Covid-19 restrictions were lifted, the move towards online education has continued to become ever more relevant, with many institutions incorporating more options for online and blended learning alongside their in-person offerings.

In addition, this growth in online learning is backed up by predictions for the future of the broader education market. For example, it was predicted that in the years between 2022 and 2030 the market for e-learning will grow by 20.5% (Hamilton & Swanston, 2024). This suggests that the expansion of online and blended learning is unlikely to stop any time soon and that investment in this field is likely to continue.

Students have also expressed a desire for more opportunities to complete their studies online. According to a 2020 study, over 73% of students reported that they would like to take an online course in the future, with 68% stating that they would be interested in taking a course which combined in-person and online instruction (McKenzie, 2020).

A survey of IT leaders in higher education found that HE institutions are beginning to recognise that digital transformation has become essential to their survival, with 67% responding that digital transformation had become a greater institutional priority and 75% responding that it would continue to grow in importance in future (Brooks & McCormack, 2020).

What approaches can institutions take to better support the teaching of practical courses online?

It is not enough for institutions to passively wait for the next emergency that will force them back into online learning. It is crucial that HE providers are proactive in equipping themselves to employ flexible models of learning design that allow for in-person, online, blended and hybrid modalities – particularly in courses with practical elements where experiential learning needs to be front and centre. In order to achieve this, institutions must invest in digital literacy development for students, professional development for staff, software, IT infrastructure, digital transformation and many other areas. However, the fundamental shift that needs to happen is a transformation of pedagogic culture.

What kind of pedagogic transformation is needed? As stated in the introduction to the book 100 Ideas for Active Learning, 'current models of education need to make radical changes… starting with developing a curriculum that supports active approaches to learning' (Gowers, Oprandi and Betts, 2022). Too many educational environments still rely on one-dimensional didactic pedagogies of the type that Paulo Freire referred to as the banking model of education, where it is assumed that knowledge can be passively deposited in the minds of students (Freire, 1970). It is time for us to move beyond these outmoded assumptions and instead place evidence-based pedagogies, such as active learning and blended learning, at the forefront of our academic communities.

What is active learning? In active learning, rather than the teacher 'transmitting' knowledge through lectures or reading, learners engage in a series of activities which require them to produce observable evidence of their learning. Where possible, these individual, pair and group tasks should aim to develop higher order thinking skills, emotional connection with content and tactile or physical engagement with the environment. Active blended learning is an extension of this that combines the principles of active learning for in-person instruction with online learning activities that emphasise student engagement, participation, and interaction.

There is overwhelming evidence on the effectiveness of active learning (e.g. Deslauriers et al., 2019; Freeman et al., 2014; Michael, 2006) and the positive impact that technology can have on learning when used appropriately (e.g. Hofer et al., 2021; Noetel et al., 2021; Bond et al., 2020). Research findings also suggest that active learning is more inclusive, having the ability to reduce awarding gaps for underrepresented groups (see Ballen et al., 2017; Theobald et al., 2020). As early as 2006, Joel Michael at Rush Medical College conducted an evaluation of whether there was sufficient evidence to support active learning and concluded that 'the very multiplicity of sources of evidence makes the argument compelling' that 'active learning, student-centred approaches…work better than more passive approaches' (Michael, 2006, p. 165). By unifying our teaching and learning cultures around active blended learning, we can ensure that learners are given the best possible chance to succeed and staff teaching practical courses online will have a more robust learning design framework in which to situate their pedagogic practice.

Even if we can change cultures of teaching and learning to recognise the effectiveness of active blended learning, how we adapt these blended and online pedagogic modalities for disciplines with substantial practical components still poses a major challenge. In general, hands-on, experiential learning is integral to practical courses, such as engineering, dance or nursing, and they often rely on physical presence in a specific location. However, innovative pedagogues around the world have demonstrated that with creativity, judicious use of technologies, and innovative teaching methodologies, practical education can often be delivered effectively online. This book explores how educators across various disciplines, and in various international contexts, have successfully navigated this transition and highlights why developing active blended learning and robust online pedagogies for practical elements of courses will be increasingly important in the future of higher education.

What is the structure of this book and what chapters are included?

To help give you an overview of the structure of this book and the chapters included, we provide a short summary below. The book is divided into five parts, with each part providing a different lens through which to view the topic of teaching practical courses online. The five parts include: student perspectives; teaching case studies; assessment; educational development; and field work and placements.

Part 1: Student Perspectives

Part 1 sets the tone for the whole publication by focusing on the perspective of the student, with contributions from an undergraduate student, a postgraduate student and the aggregated views of many students through survey responses.

Chapter 1: My experience of practical engineering labs as a first-year undergraduate during the pandemic. Author: Thanassis Frank, Trinity Hall, University of Cambridge.

In Chapter 1, Frank presents a reflective narrative on his experience as a first-year undergraduate engineering student during the Covid-19 pandemic. The chapter discusses how practical labs, crucial for hands-on learning, were adapted to an online format. Frank details three key lab experiences—Electronics, CAD, and Structures—where he learned to engage with remote tools and simulations effectively. The chapter delves into his sense of isolation and the challenges of lacking hands-on experience during the pandemic. Despite the challenges, these experiences were enriched by innovative approaches, such as using personal kits and virtual simulations. As Frank reflects, 'this independent, task-based learning engaged me meaningfully with the equipment provided, and I gained experience using it'. The chapter underscores the potential for blended and online labs in the future, suggesting that such methods could enhance flexibility and collaboration in engineering education.

Chapter 2: Connection loss in online learning: a physiotherapy student's experience of practical education during a pandemic. Authors: Dr. Laura Blackburn, Dr. Larissa Kempenaar, and Dr. Sivaramkumar Shanmugam, Glasgow Caledonian University.

In Chapter 2, Blackburn, Kempenaar, and Shanmugam discuss the shift from hands-on to online learning in physiotherapy at Glasgow Caledonian University, exploring its impact on skill acquisition and peer interaction. The chapter highlights challenges like reduced hands-on practice and social interaction, which affected student engagement and skill development. It also emphasises the necessity of fostering social engagement in virtual environments and integrating cognitive, psychomotor, and affective learning in hybrid settings. The authors address the psychological impact of the pandemic, stressing the importance of mental health support in remote learning scenarios. In addition, they suggest ways to enhance virtual practical sessions to better simulate physical interactions, concluding that 'the inherent flexibility of hybrid approaches and refocusing of priorities promoted equity and enhanced students' learning and well-being'.

Chapter 3: Online versus live practicals: what were physiology students missing during the pandemic? Authors: Professor Matthew J. Mason and Dr. Kamilah Jooganah, University of Cambridge.

In Chapter 3, Mason and Jooganah examine the experiences of physiology students at the University of Cambridge who transitioned from in-person to online practical classes. The chapter compares student perceptions of online and live practicals, emphasising the importance of hands-on learning and the challenges of replicating this experience virtually. While students appreciated the interactive elements of online classes, they generally did not view them as adequate replacements for in-person practicals. The chapter highlights the value students placed on active learning, with 'working in an "active" way' receiving the highest rating among 17 aspects of practicals. The authors recommend incorporating hands-on activities that can be safely performed at home and enhancing online collaborative learning to improve the effectiveness of online practical education. They reflect on the importance of hands-on experience for physiology students and suggest improvements for future online practical courses. Their findings underscore the need for more interactive and engaging online tools to more closely approximate the in-person practical experience.

Part 2: Teaching Case Studies

Part 2 provides a range of case studies presented from the perspective of the teacher. These span a range of disciplines, from chemistry to performing arts to business negotiation skills.

Chapter 4: STEM: Teaching practical analytical chemistry online: improving delivery of a Year 2 NMR spectroscopy practical. Authors: Dr. Cate Cropper and Prof. Gita Sedghi, University of Liverpool.

In Chapter 4, Cropper and Sedghi detail the transition of analytical chemistry labs to an online format at the University of Liverpool, utilising remote access to spectrometers and enhanced digital feedback mechanisms. In addition to these solutions, the online format also improved students' access to demonstrators and fostered peer collaboration through breakout rooms and Microsoft Teams, which simulated the lab environment. The authors observed that 'students were confident in explaining concepts to each other and worked together after building a rapport with other members of their group', highlighting the benefits of active, collaborative learning even in a virtual setting. They reflect on these unexpected improvements in student engagement and performance, discussing how these online methods provided a community-like atmosphere similar to in-lab experiences. The authors also explore the challenges and solutions involved in maintaining academic rigour in an online setting, proposing the potential for these adaptations to be applied more broadly across scientific disciplines.

Chapter 5: Digital technologies and practices for the online teaching of practical skills in performing arts disciplines within UK conservatoires. Author: Evan Dickerson, Guildhall School of Music & Drama.

In Chapter 5, Dickerson explores the use of digital technologies in performing arts education at the Guildhall School of Music & Drama, detailing the application of online meeting software such as Zoom and Moodle. He highlights the unique challenges of teaching music performance online, discussing the benefits of video-based learning using platforms such as eStream, which make it 'possible for musicians to rehearse and perform orchestral repertoire whilst socially distanced across multiple spaces simultaneously…watching streamed feeds of a conductor and their distributed colleagues'. Dickerson examines how these digital adaptations can complement traditional teaching methods, enhancing the learning experience for students, and provides insights into how educators can foster creativity and collaboration in a virtual environment.

Chapter 6: Zoom-based negotiation – a path to tacit understanding. Scaffolding employability within the undergraduate curriculum. Authors: Rebecca Payne and Joe Bramall, Harper Adams University.

In Chapter 6, Payne and Bramall examine how business negotiation skills were taught online using Zoom at Harper Adams University. The chapter details how traditional face-to-face classes were successfully adapted to Zoom, enabling students to hone their negotiation skills in a virtual setting. This transition preserved learning effectiveness and introduced new advantages, such as reducing performance anxiety and fostering engagement through random pairings in breakout rooms. The online meeting format allowed for efficient role allocation, observation, and instant feedback, which were challenging in a physical classroom. The chapter highlights the value of active learning and reflection in skill development, with Bramall noting: 'recording progress within an ePortfolio…allowed a more detailed reflection of performance, again furthering the ability to develop my skills'. This approach helped students build confidence and showcased the potential of digital tools in enhancing employability.

Part 3: Assessment

Part 3 provides an opportunity to focus on assessment, with chapters on incorporating industry connections in assessment, redesigning nursing assessments for online learning, and using virtual reality to assess learning in creative disciplines.

Chapter 7: Hands-on assessment in a hands-off world: strategic implementation of inclusive assessment in an online environment. Authors: Professor Sarah Jones, Southampton Solent University, and Professor Alasdair Blair, De Montfort University.

In Chapter 7, Jones and Blair examine how the Covid-19 pandemic accelerated the shift towards more inclusive, flexible, and innovative assessment methods in higher education. Taking media production courses at De Montfort University (DMU) as a case study, the chapter details how traditional hands-on assessments were adapted to online formats, ensuring that students could still meet learning outcomes despite limited access to high-end equipment. For example, students were encouraged to use mobile phones for their projects, emphasising that 'quality came from knowing how to use the tools rather than the quality of the tools themselves'. This approach ensured fairness and maintained high standards, regardless of students' technological resources. The chapter advocates for the continued evolution of assessment methods to incorporate the lessons learned during the pandemic, emphasising the need for flexibility, creativity, and a focus on student-driven learning in a digital age.

Chapter 8: Covid-19 lockdowns: a catalyst for rethinking assessments in skill-based nursing courses. Authors: Dr. Zeenar Salim, University of Georgia, Dr. Anil Khamis, Shanaz Cassum, Zohra Kurji, the Aga Khan University, and Professor Pammla Petrucka, University of Saskatchewan (USask) and the Aga Khan University.

In Chapter 8, Salim, Khamis, Cassum, Kurji, and Petrucka explore how the Covid-19 lockdowns acted as a catalyst for rethinking assessments in skill-based nursing courses at the Aga Khan University School of Nursing and Midwifery in Pakistan (SoNAM-PK). Faced with challenges such as unreliable internet access and cultural restrictions, faculty had to innovate quickly, moving from traditional face-to-face assessments to online formats. Various assessment tools, such as virtual simulations and case-based online viva exams, were introduced. As one faculty member noted, 'This opportunity of lockdown made us think differently, redesign our assessments and conduct them successfully'. The chapter highlights the importance of collaboration and empathy in creating equitable and effective assessments, with lessons learned that could influence future educational practices.

Chapter 9: Creative performing arts: ways of witnessing practice. Author: Beth Loughran, University of Cumbria.

In Chapter 9, Loughran examines the adaptation of assessment methods for practical dance education during the pandemic. The chapter outlines how traditional assessments were rethought when students were confined to home studios, dealing with challenges like limited space and camera constraints. It explores alternative assessment methods, such as critical accounts of practice, to ensure fairness and equity. The author reflects that the 'idea was to collect critical accounts of practice through documentary methods (oral presentation, documentation of practice, written accounts), in place of bodily practical ones (performance of dance)' to create a more holistic assessment of dance performance along with the use of video conference calls. The chapter also emphasises student perspectives and underscores the importance of access and inclusivity, advocating for innovative and equitable assessment approaches within higher education. As Loughran concludes, 'despite the challenges, the act of training, crafting, creating, and performing ensued, proving that the essence of dance can adapt and thrive in new environments'.

Part 4: Educational Development

Part 4 shifts perspectives to reflect on educational development and approaches for developing teaching quality across an institution, featuring chapters on playful learning and escape rooms, guidance for facilitating highly collaborative workshops online, and a framework for supporting teaching staff to develop their skills in building communities when transitioning to online teaching.

Chapter 10: Playful practicals: breaking free from educational norms with online escape rooms. Authors: Dr. Emma Gillaspy, Dr. Abhi Jones, Gemma Spencer, and Julia Robinson, University of Central Lancashire.

In Chapter 10, Gillaspy, Jones, Spencer, and Robinson discuss the innovative use of online escape rooms in healthcare education at the University of Central Lancashire. The chapter discusses how traditional hands-on learning experiences were adapted into digital formats, allowing students to develop critical skills like teamwork, communication, and leadership in a playful yet challenging environment. The escape rooms facilitated active learning by requiring students to solve clinical puzzles collaboratively, reflecting real-world healthcare scenarios. The authors highlight the importance of allowing students to explore and solve problems independently, with one noting, 'Letting go of the need to be the expert has been a lightbulb moment for each of us, allowing students to find their own path through the learning experience'. This approach not only engaged students in meaningful learning but also prepared them for the complexities of modern healthcare, emphasising creativity and self-determined learning. The success of these escape rooms has inspired a broader application across the university, promoting a culture of playful and active learning.

Chapter 11: Doing what it says on the tin, online: the Workshop module for Open Education practitioners. Authors: Professor Chrissi Nerantzi, University of Leeds, and Dr. Javiera Atenas, University of Suffolk

In Chapter 11, Nerantzi and Atenas discuss their experience designing and delivering an online Workshop module for the Masters in Open Educational Leadership at the University Nova Gorica. The module aimed to empower students, who were open educators from Europe and Africa, by facilitating deep, hands-on engagement through digital platforms. The pedagogical approach focused on peer-to-peer learning, project-based activities, and creating a strong learning community. Live sessions, structured around key stages of project development—Introducing, Exploring, Designing, Testing, and Evaluating—guided students through their individual open education projects. Despite technical challenges, the module successfully fostered a collaborative environment where students felt motivated and connected. As the authors highlighted, 'students learnt experientially through making, on and off screen, leading to the creation of authentic projects which they planned to apply in real life'. The chapter underscores the potential of online practical modules to create meaningful and impactful learning experiences, which can be adapted to larger cohorts and various disciplinary contexts.

Chapter 12: Undoing learning through lurking: a simple framework for supporting staff thinking about online teaching. Author: Charlie Reis, Xi'an Jiaotong-Liverpool University (XJTLU).

In Chapter 12, Reis discusses the challenges of engaging students in online learning environments, including how to address 'lurking,' where students passively observe without active participation. Drawing on experiences at Xi'an Jiaotong-Liverpool University (XJTLU), Reis introduces the 3-L framework—Listening, Learning, and Leveraging—as a strategy to combat this passivity. The framework aims to progressively increase student engagement by starting with basic tasks that evidence memory (Listening), moving to deeper understanding and connection of concepts (Learning), and culminating in activities that require students to apply their knowledge in meaningful ways (Leveraging). The chapter emphasises the importance of creating structured opportunities for interaction and making learning visible to instructors. Reis asserts that 'getting students to present/represent themselves online in a structured way…makes a lot of sense for online learning' and highlights the need for thoughtful design in online education to ensure active participation and deeper cognitive engagement.

Part 5: Field Work and Placements

Part 5 provides an opportunity to look at two aspects that are often overlooked in online learning: field work and placements. In venturing beyond the classroom, we have the chance to see an example of how to create virtual fieldwork experiences, as well as a chapter on helping learners gain valuable real-world experience in the workplace by conducting virtual placements.

Chapter 13: Moving away from the traditional PowerPoint to an interactive documentary-style session. Author: Dr. Andrew Watson, Harper Adams University.

In Chapter 13, Watson explores the transition from traditional PowerPoint lectures to an interactive documentary-style format during the Covid-19 pandemic. The shift was